ECEN 3730: Board 2 Report
Switching Noise with Good and Bad Layout

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10/16/26

Objective:

The objective of this PCB design project was to design, assemble, and test two electrically identical hex-inverter circuits with different PCB layouts, driven by a common 555 timer operating at approximately 500 Hz. The good layout used a continuous ground plane and a decoupling capacitor placed close to the inverter's power pin. The bad layout used ground traces instead of a continuous ground plane and placed the decoupling capacitor farther from the IC. Test points were included to measure switching noise on the quiet-HIGH outputs, quiet-LOW outputs, and board-level power rail. Comparing these measurements allowed the effects of the layout on switching noise to be evaluated.

Plan of Record (POR):

The plan for Board 2 is to compare switching noise in two electrically identical hex-inverter circuits using good and bad PCB layouts. A 5 V DC input will supply an LDO that generates 3.3 V for the 555 timer and inverter ICs. The timer will generate an approximately 500 Hz, 60% duty cycle signal. The good layout will use a continuous ground plane and nearby decoupling, while the bad layout will use ground traces and a faraway decoupling capacitor.

Each hex IC will have three outputs driving LEDs through 47 Ω resistors, one unloaded switching output, one quiet-HIGH output, and one quiet-LOW output. Isolation switches and pull-up resistors will keep inputs defined when disconnected from the timer. Labeled test points and indicator LEDs will also be included for debugging and testing.

Successful operation will require correct supply voltages, expected timer timing, inverted outputs, working LEDs, and functioning switches. Testing will measure rise time, LED current, and good/bad switching noise using 10× probes with ground springs. Additional tests will estimate inverter output resistance and examine LDO behavior with and without its output filter capacitor.

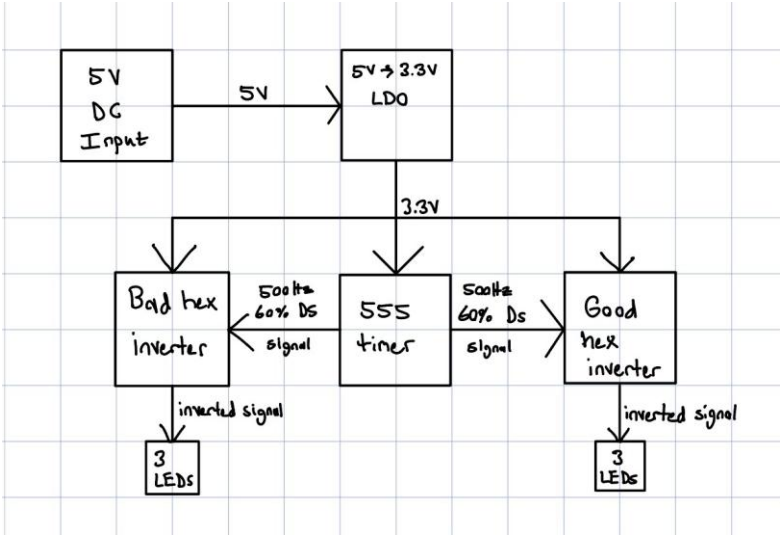
Key Components:

LMC555 (555 Timer) , SN74AHC14 (Hex Inverter ICs), AMS1117-3.3 (5 V to 3.3 V LDO)

Requirements for Board to work:

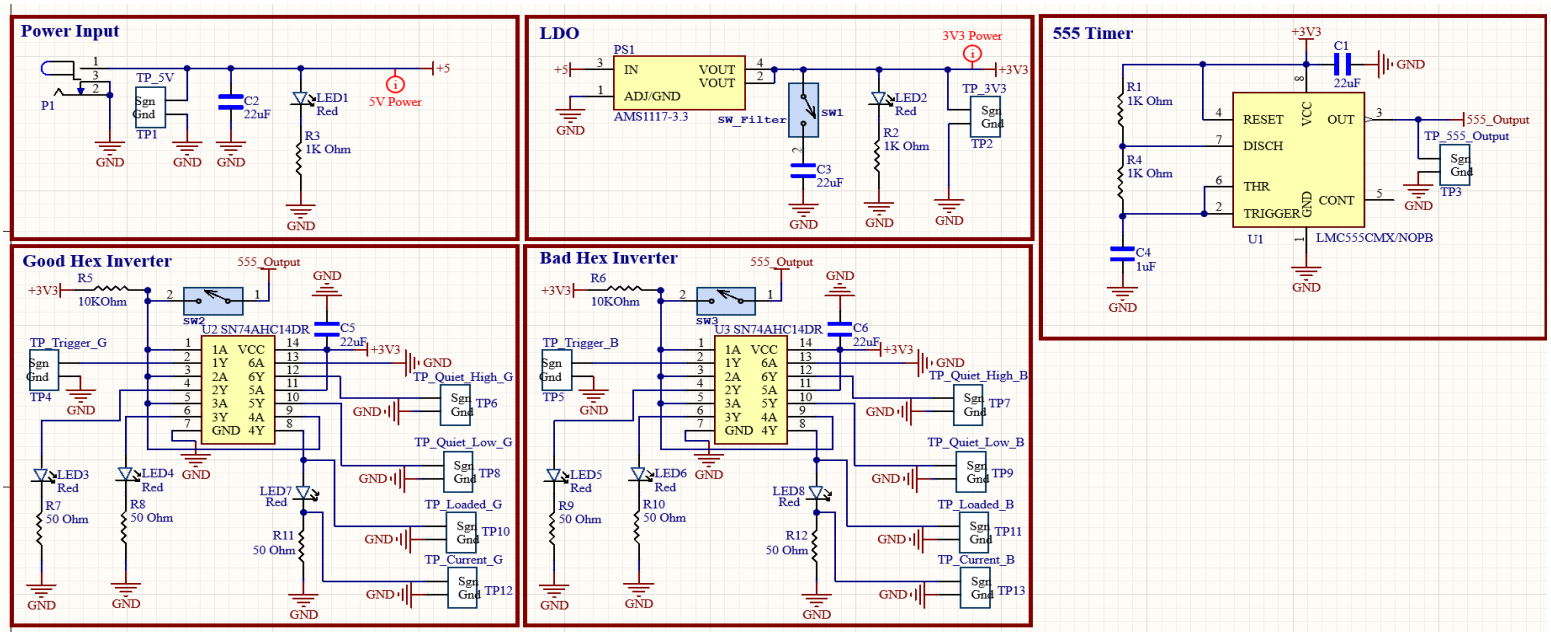
- Input rail measures approximately 5V, and the LDO output measures about 3.3V
- Power indicator LEDs turn on when board is powered on
- The 555 timer produces an approximately 500 Hz square wave, 60% duty cycle
- Both hex inverter circuits produce outputs opposite to the timer signal
- Isolation switches disconnect the intended circuit sections
- Quiet High output measure around 3.3V, and quiet Low outputs measure near ground between switching
- Test points allow measurements of power rail switching noise

Sketch and Calculations:

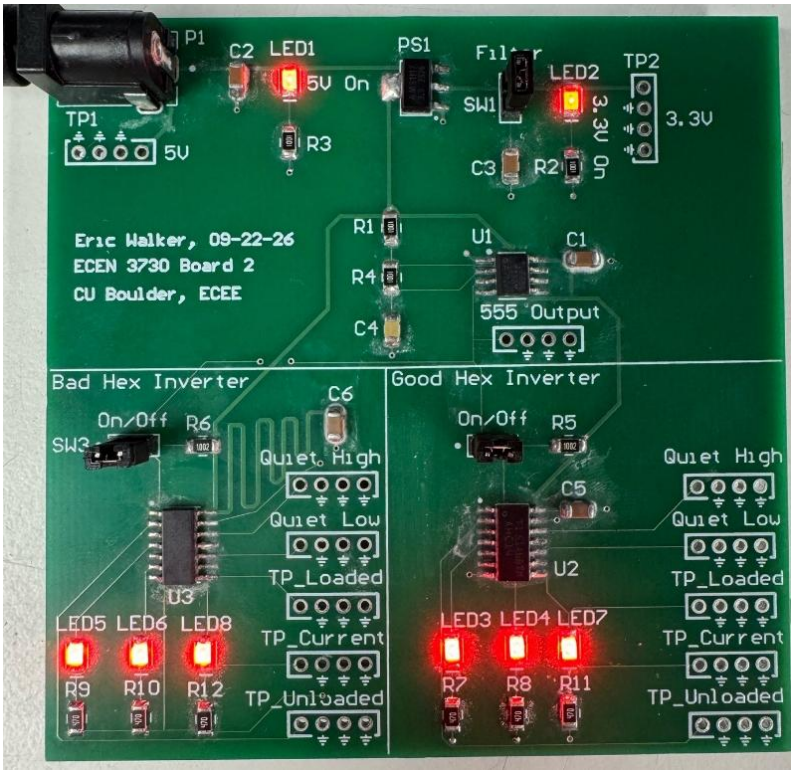


Current Through LED's: $I = \frac{V}{R} = \frac{0.7}{47} = 0.0149\ A = 14.9\ mA$

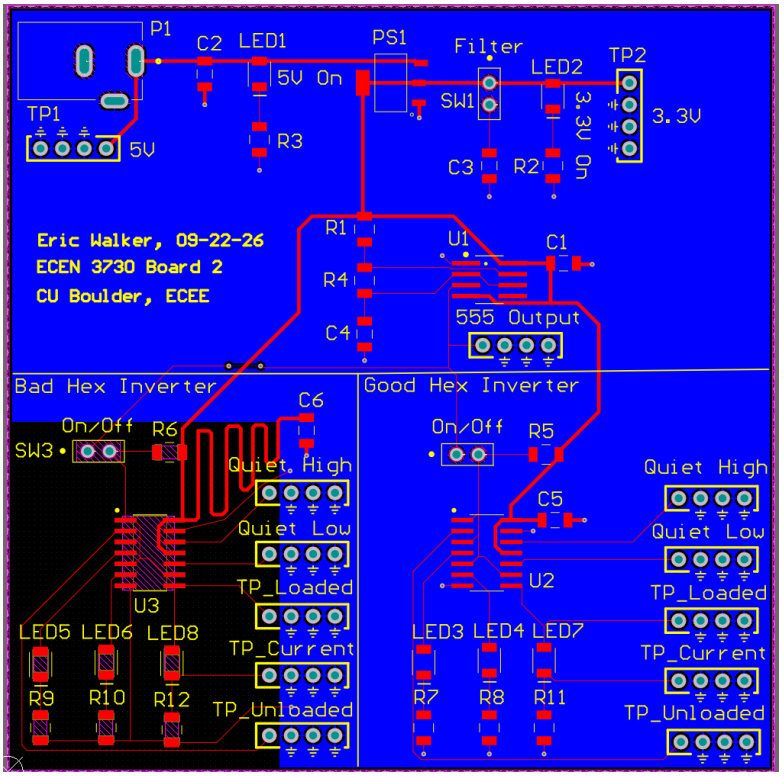
Schematic:



Board Implementation:



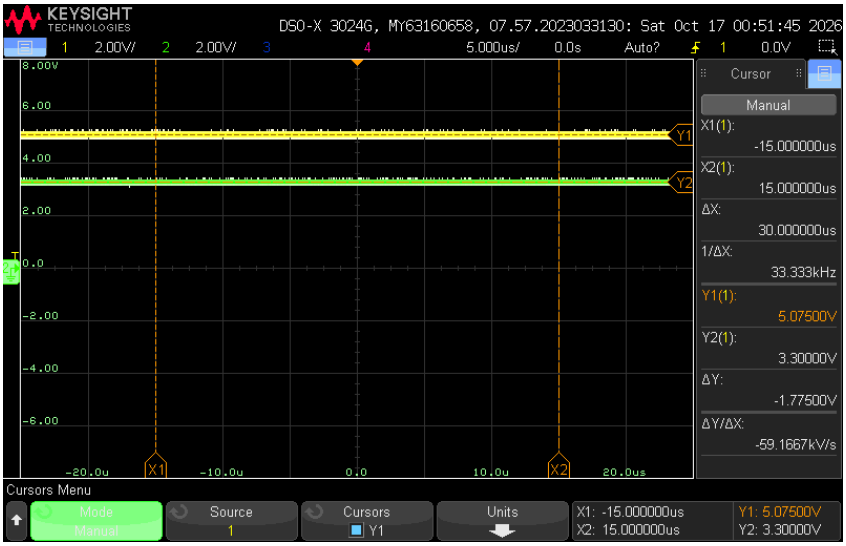
Layout:



Scope Shots and Analysis:

1. <u>Power Rail Verification:</u>

Figure 1: Power Rail Measurement

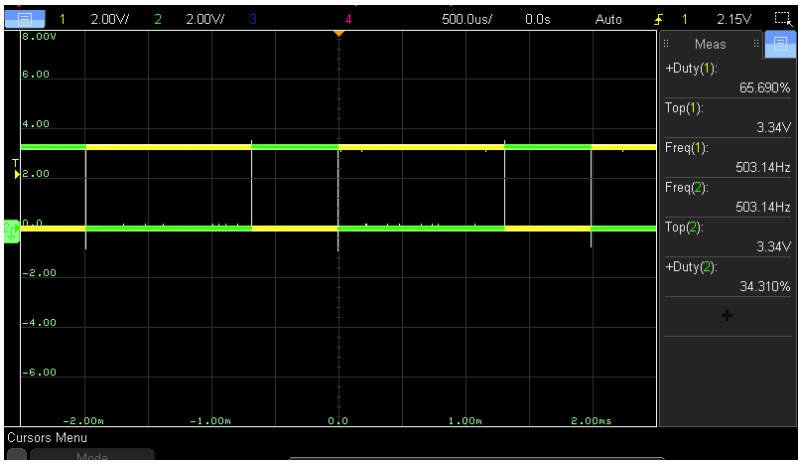


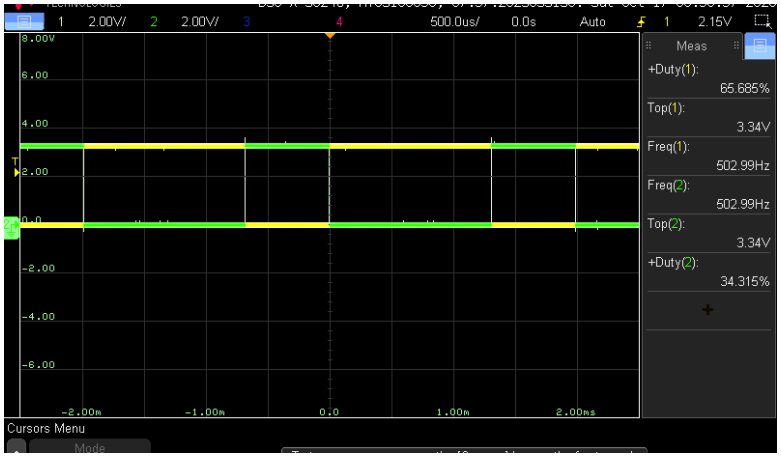
The 5 V input rail is shown in yellow, and the regulated 3.3 V rail is shown in green. Cursor measurements indicate approximately 5.075 V at the barrel-jack output and 3.300 V at the LDO

output. These measurements confirm that the power circuit provides the expected DC supply voltages.

2. Timer and Hex Inverter Operation:

Figures 2 and 3: Timer and inverter output waveforms for bad(left) and good(right) layouts

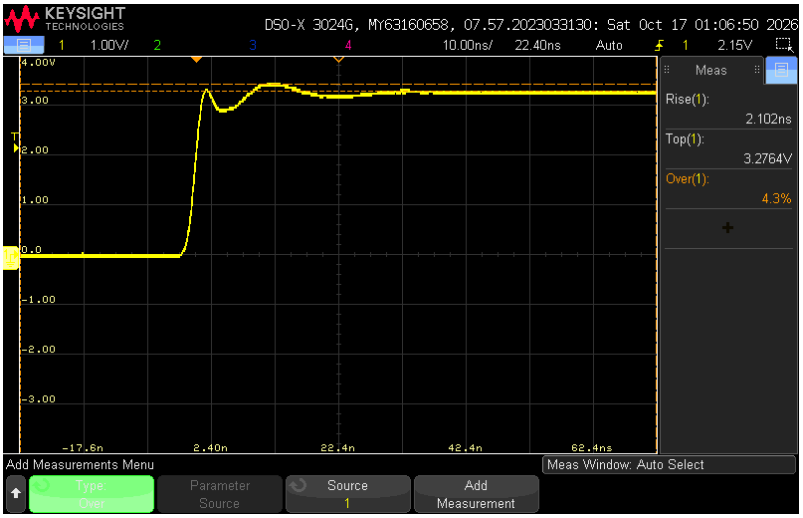


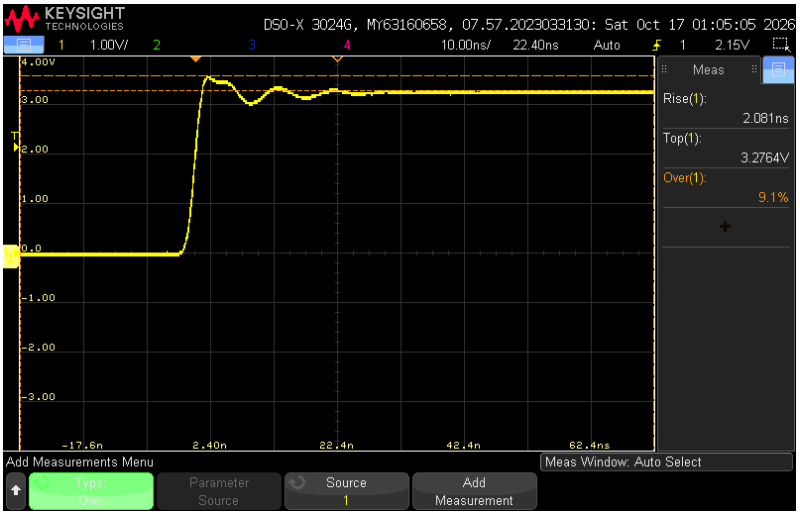


The measurements taken at the 555-timer output as well as the unloaded hex inverter outputs show the desired operating behavior of the inverter directly inverting the 555-timer output with matched frequencies, high voltages, as well as the positive duty cycle measurements of each adding to exactly 100%. This verifies the inverter circuits on both layouts are working properly.

3. Unloaded Inverter Rise Time

Figures 4 and 5: Rising Edges of Unloaded Inverter Outputs, bad(left), good(right)

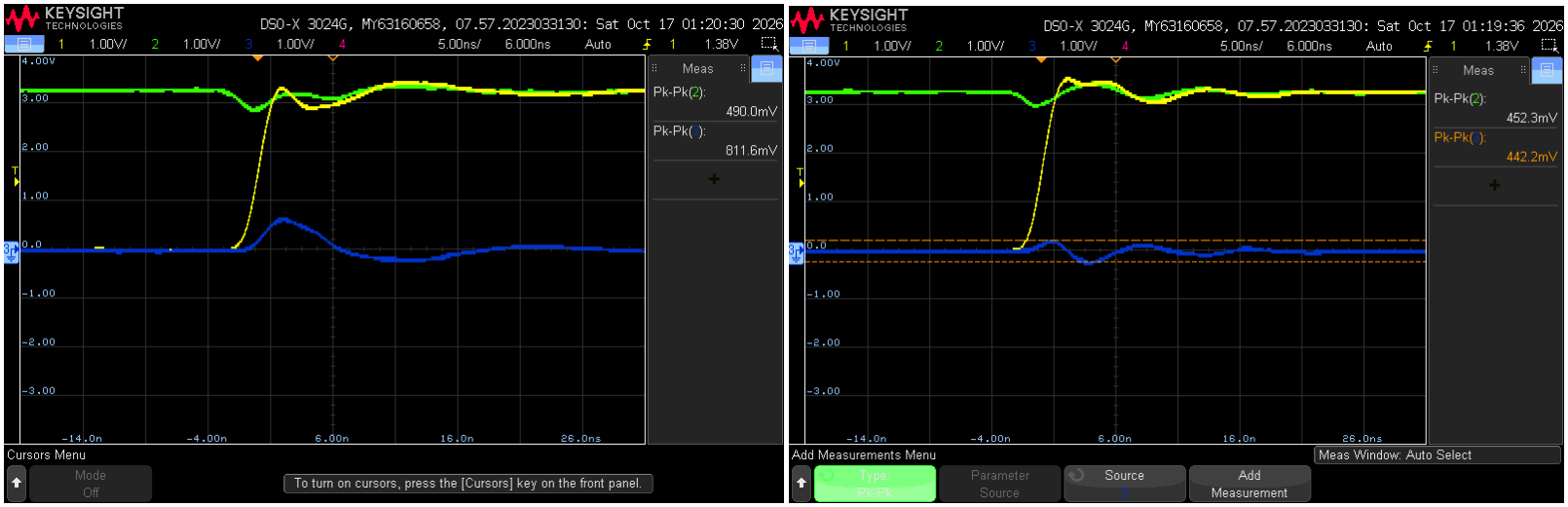




The unloaded hex-inverter outputs had nearly identical measured rise times of approximately 2.1 ns and output-HIGH voltages of approximately 3.28 V for both layouts.

4. Quiet-High and Quiet-Low Switching Noise

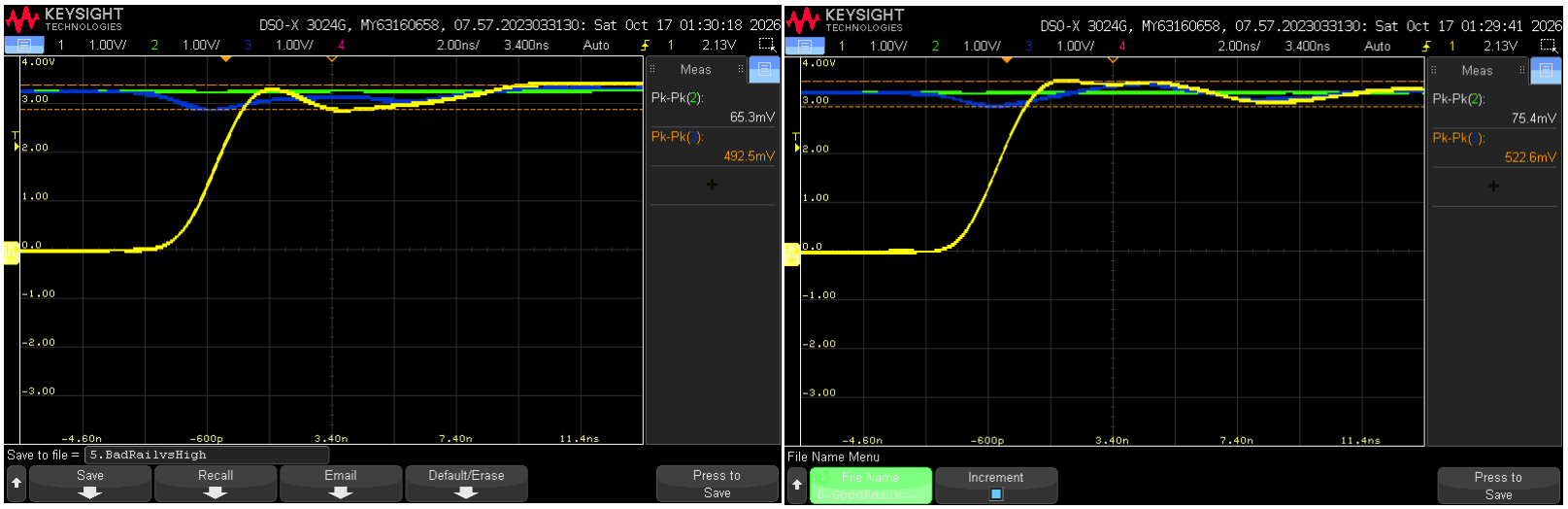
Figures 6 and 7: Quiet-HIGH and quiet-LOW switching noise for the bad layout (left) and good layout (right).



The quiet-HIGH (green) and quiet-LOW (blue) outputs were measured while triggering on the rising edge of the unloaded inverter output (yellow). Although these quiet outputs do not intentionally switch, both showed disturbances coinciding with the switching event. The quiet-HIGH noise measured 452.3 mVpp for the good layout and 490.0 mVpp for the bad layout. Quiet-LOW noise showed a larger difference: 442.2 mVpp versus 811.6 mVpp, approximately 46% lower on the good layout. This is consistent with the continuous ground plane reducing return-path inductance and ground bounce.

5. Board Level Rail Noise Compared with Quiet High

Figures 8 and 9: Board level 3.3V rail and quiet HIGH switching noise for good and bad layout

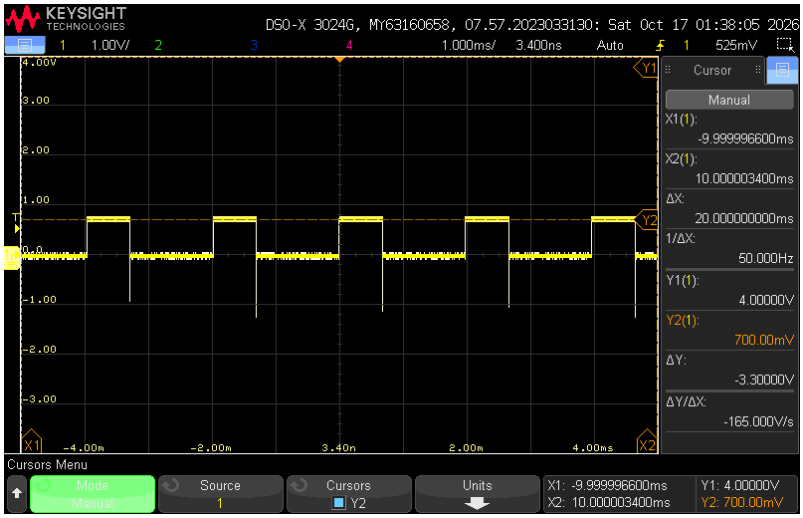


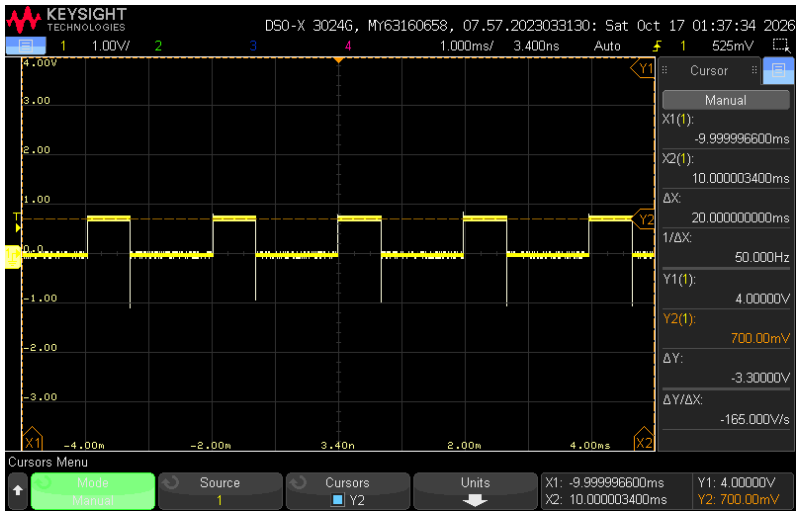
The board 3.3V rail (green) measured 75.4 mVpp for the good layout and 65.3 mVpp for the bad layout, while quiet HIGH (blue) measured 522.6 mVpp and 492.5 mVpp, respectively. Both

layouts showed substantially greater noise on quiet HIGH than on the board rail. This reflects the interconnect and package impedance between the board supply and the IC's internal supply. These measurements show that the quiet-HIGH output has substantially greater switching noise than the board-level power rail.

6. LED Load Current

Figures 10 and 11: Voltage across a 47 Ω LED series resistor for the bad layout (left) and good layout (right).



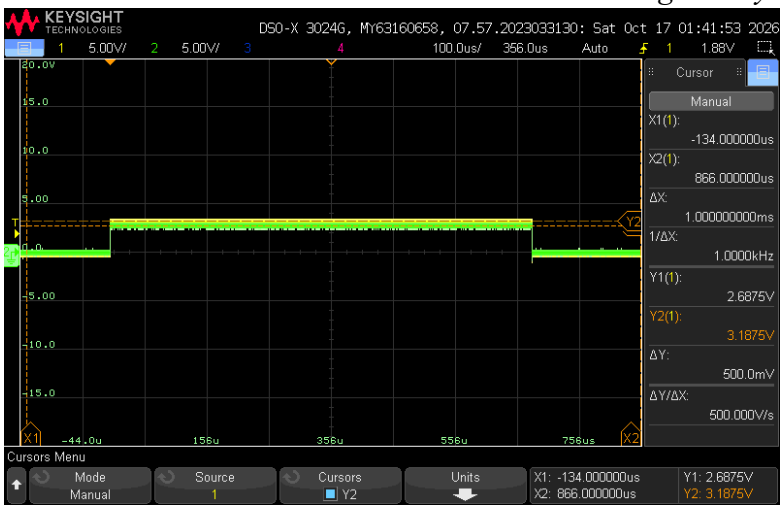


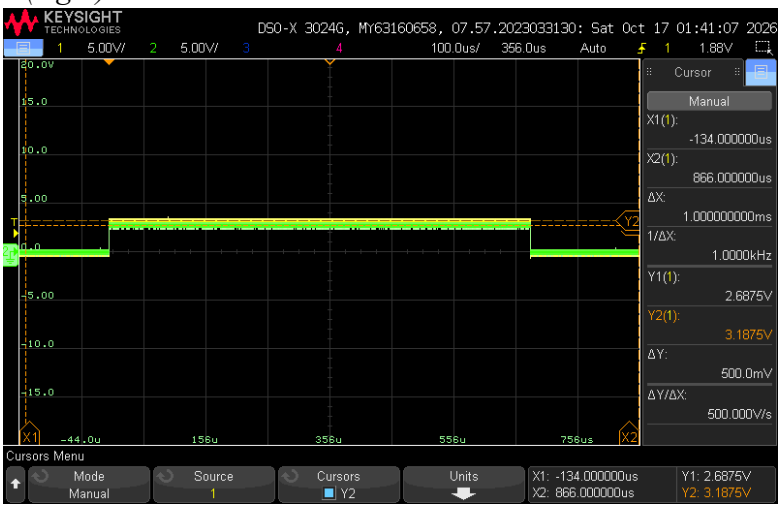
The voltage across the 47 Ω series resistor measured approximately 0.700 V in both layouts. Since the LED and resistor are in series, the LED current can be calculated using Ohm's law:

$$I_{LED} = \frac{0.7\ V}{47\ \Omega} = 0.0149\ A = 14.9\ mA$$

7. Hex Inverter Output Resistance

Figures 12 and 13: Loaded and unloaded inverter output voltages for the bad layout (left) and good layout (right).





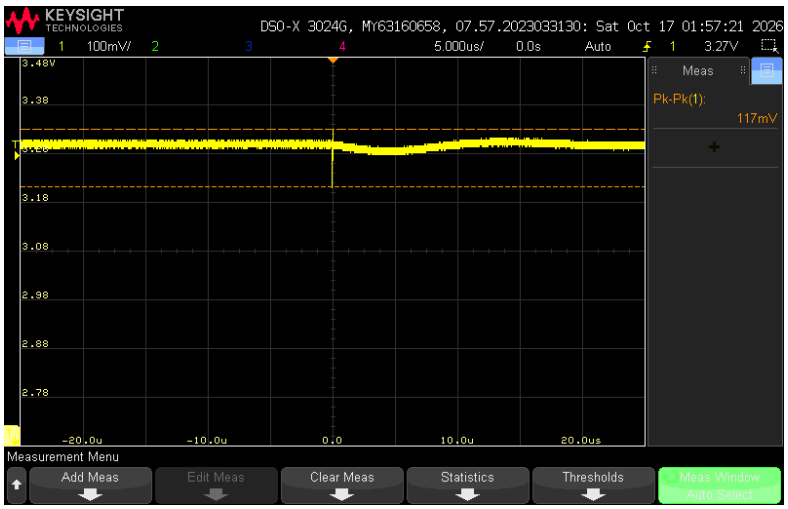
Both layouts showed an unloaded output-HIGH voltage of approximately 3.1875 V and a loaded output-HIGH voltage of approximately 2.6875 V. Using the LED current calculated previously, the effective output resistance was estimated as:

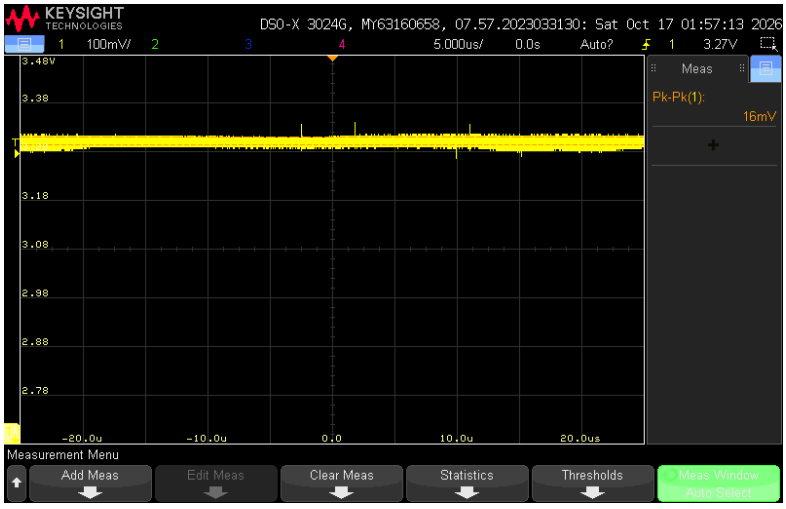
$$R_{TH} = \frac{(V_{Unloaded} - V_{Loaded})}{I_{LED}} = \frac{(3.1875 - 2.6875)V}{0.0149\ A} = 33.6\ \Omega$$

The approximately 0.500 V drop under load demonstrates that the inverter output has finite resistance. Both layouts produced near identical estimates at this operating point.

8. Effect of the LDO Output Filter Capacitor

Figures 14 and 15: LDO output voltage with the output filter capacitor disconnected (left) and connected (right).





The LDO output measured 117 mVpp with the output filter capacitor disconnected and 16 mVpp with it connected. Both captures used the same 5 µs/div time scale. The much larger Vpp measurement on the left is mostly due to a large voltage spike as seen in the measurement but the small ripple in the voltage shows how the power rail is affected without the capacitor. When adding the capacitor, we see a much more stable 3.3 V rail which is the result of having a local capacitor directly on the output of the LDO which stabilizes the voltage and gives a more consistent voltage underload.

Project Analysis:

Effect of Layout Switching Noise:

The clearest benefit shown of the good layout compared with the bad layout was much lower quiet-LOW noise. The good layout measured 442.2 mVpp, compared with the 811.6 mVpp on the bad layout, a reduction of 46%. Quiet-HIGH noise showed a smaller difference, measuring 452.3 mVpp for the good layout and 490 mVpp for the bad layout. These results are consistent with the continuous ground plane reducing return path inductance and ground bounce. The unloaded inverter rise times were nearly identical: 2.081 ns for the good layout and 2.102 ns for the bad layout.

The 3.3 V board rail showed relatively small switching disturbances, 75.4 mVpp on the good layout and 65.3 mVpp for the bad layout, showing that the switching noise on the power rail was very low. The quiet-HIGH output showed much larger disturbances, measuring 522.6 mVpp for the good layout and 492.5 mVpp for the bad layout. Although quiet HIGH is held near the supply voltage, it is driven by the internal IC circuit. Impedance in the PCB connections and IC package allows that internal voltage to fluctuate during switching even when the board rail stays relatively stable.

What Worked and Should be Repeated:

The scope measurements verified the expected supply voltages and correct operation of the circuits such as the 555 timer and the hex inverters. Including well labeled test points as well as switches to each separate circuit and indicator LEDs all allowed easy testing and debugging. It was clear from the measurements the benefits of the good design practices used on the good layout, such as the significantly lower quiet-LOW noise on the good layout. For my future boards, I will use these good practices like a continuous ground plane, and close decoupling capacitors.

Hard Errors and What Didn’t Work:

After receiving my board and assembling it using all the correct resistor and capacitor values, I had no issues getting it to work properly. By starting my design early, spending time double checking my schematic and layout, and having my design done by CDR, I was able to have a fully functional board with no errors. Thankfully I did not need to perform any debugging and every piece worked as intended which made testing and measuring very easy.

Soft Errors and What to do Different Next Time:

There were no soft errors I experienced on my physical board, but I believe I could have done a better job getting clearer and more consistent scope measurements. Some of my scope measurements were hard to read and after looking back at them, next time I will use a greater vertical scale to make the small disturbances more visible. For future boards, I will learn from this one and use continuous ground planes and minimize power and return loops by placing close

decoupling capacitors close to the power pins and close ground vias. The results from the board show the physical benefits of these good design practices.

Conclusion:

Board 2 was successfully assembled and tested, with both hex-inverter circuits operating at approximately 503 Hz driven by the 555 timer. The good layout showed approximately 46% less quiet-LOW noise, while the quiet-HIGH results showed a smaller difference. Measurements also demonstrated that quiet-HIGH noise can be substantially greater than board level supply noise. These results support using continuous ground planes, short power and return paths, and nearby decoupling capacitors to reduce switching disturbances. The project also reinforced the importance of including many test points, on and off switches, and consistent probing when comparing PCB performance. From this board, I was able to see the real benefits of using continuous ground planes and decoupling capacitors which were able to be classified in real measurements comparing two identical circuits.